DeepDiff-SHAP: Interpretable deep learning for subgroup-specific causal hypothesis generation using conditional SHAP

Aditya Sriram[1], Soyeon Kim[2], Joseph A Carcillo[2], Hyun Jung Park[1]
[1]Department of Human Genetics, University of Pittsburgh, Pittsburgh, PA, USA
[2]Department of Pediatrics, University of Pittsburgh, Pittsburgh, PA, USA

Motivation

- As precision medicine continues to gain traction, there is a growing need for analytical methods that can identify subgroup-specific risk factors that either influence disease risk or therapeutic responses differently across distinct populations, or differently across multiple states within the same population
- Traditional causal inference frameworks, however, typically estimate average treatment or exposure effects across a global population grouping

Project Description

- We introduce DeepDiff-SHAP as a principled, statistically sound three-algorithm approach rooted in regression and deep learning (DL) frameworks for causal structure changes between two states
- DeepDiff-SHAP integrates conditional Shapley Additive exPlanations (SHAP) to estimate conditional dependencies and perform nonlinear differential causal inference

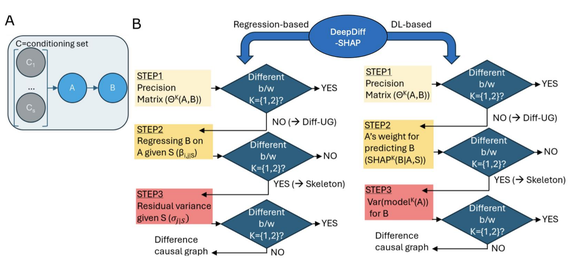


Figure 1: Method design comparing a regression-based approach to a DL-based approach.

Context

- In a standard two-group approach, causal models are estimated independently for each group (e.g., diseased and healthy)
  - differences in effect estimates are compared post hoc
  - does not account for estimation variance statistical significance of the differences
- By integrating theoretical rigor of differential causal inference with the interpretable power of deep learning and SHAP, we can discover meaningful changes in causal structure

University of Pittsburgh

Deep learning-based differential causal inference to detect **changes in causal relationships** across patient subgroups.

Precision medicine aims to tailor healthcare strategies to individual differences in genetic, clinical, and environmental factors. However, identifying **subgroup-specific causal relationships** in complex biomedical data remains a major challenge, especially when standard causal inference methods average over population heterogeneity. We introduce **DeepDiff-SHAP**, a novel framework that combines regression-based and deep learning-based differential causal inference to detect changes in causal relationships across patient subgroups. DeepDiff-SHAP integrates **conditional SHapley Additive exPlanations (SHAP)** to estimate conditional dependencies and perform nonlinear differential causal inference in a principled, interpretable manner. Applying DeepDiff-SHAP to two population-scale datasets, the **CDC Diabetes Health Indicators Dataset** and a **UK Biobank** sepsis cohort stratified by hypertension status, we identified clinically meaningful and subgroup-specific causal changes in relationships between features across the datasets including age, general health, alkaline phosphatase, and cholesterol. Our results reinforce the idea that **deep learning enhances sensitivity to complex interaction patterns** overlooked by linear models, providing new biological insights into disease progression and comorbidity-specific risk mechanisms. DeepDiff-SHAP offers **a scalable and interpretable solution** to uncover individualized causal pathways, advancing the goal of truly personalized medicine.

Step 1: Identify Differential Graph Dependencies (Precision Matrix)

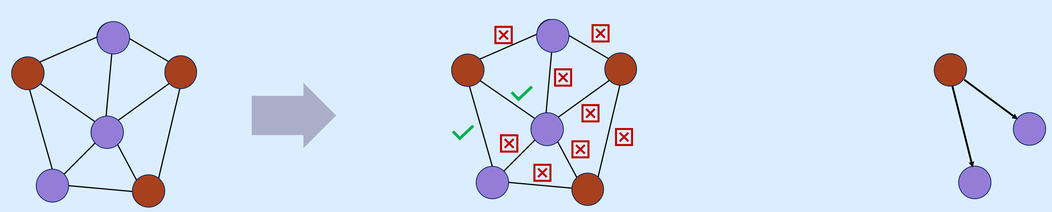
- *Identify candidate node pairs whose conditional dependencies differ between states, using changes in the precision matrix (pseudoinverse of covariance)*

Step 2: Prune Stable Relationships (Conditional Invariance Test) via SHAP

- *Test whether dependency strengths remain invariant across conditions after conditioning on neighboring nodes; prune edges showing SHAP invariance*

Step 3: Infer Directionality (DNN Residual Variance Test)

- *Estimate residual variances with deep neural networks (DNNs); directional edges are inferred when residual invariance identifies causal parents.*



Difference undirected graph (Δ-UG) → **Prune Δ-UG and identify causal skeleton** → **Orient edges and assign directionality**

Results – CDC Diabetes Data

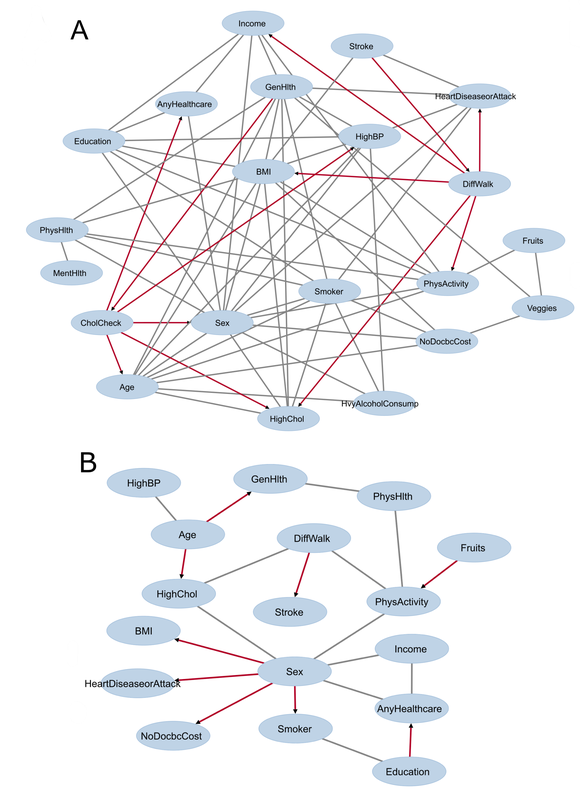


Figure 2: Differential causal inference graph on the CDC diabetes data as identified by (A) the regression-based and (B) deep-learning-based DeepDiff-SHAP. Gray lines represent the association changes that are not attributed as differential causal relationships and red arrows represent the differential causal relationships.

Interpreting Causal Graphs

- Age → high cholesterol and sex → coronary heart disease or myocardial infarction
  - have been previously mentioned in the context of type 2 diabetes being associated with women having a greater risk of cardiovascular disease, as well as studies implicating earlier cardiovascular disease events in women with T2D compared to men
- These findings suggest that deep learning-based causal inference can uncover subtle and nonlinear changes in causal structure that may be missed by linear models.

Precision Medicine Impact

- DeepDiff-SHAP's outputs support **clearer communication of subgroup-specific risks** and help prioritize the most important risk factors for patient care.

References

- Belyaeva, A., Squires, C. & Uhler, C. DCI: learning causal differences between gene regulatory networks. Bioinformatics 37, 3067-3069 (2021)
- Scott M Lundberg, S.-I. L. in International Conference on Neural Information Processing Systems Vol. 30 (Advances in Neural Information Processing Systems, Long Beach, CA, 2017)
- Lundberg, S. M. & Lee, S.-I. Vol. 30 (eds I. Guyon et al.) (2017)

Thank you to the Pacific Symposium on Biocomputing (PSB) 2026 for the opportunity to share our work.